

The weak octonionic Stiefel space $V_2^w(\mathbb{O}^2)$ is not a manifold

A full negative answer to modified James Question (2) in arXiv:2605.07226

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Abstract

Huo, Ren, and Xu ask whether their weak octonionic Stiefel spaces $V_k^w(\mathbb{O}^n)$ are manifolds. We give a negative answer in the smallest nontrivial case: $V_2^w(\mathbb{O}^2)$ is not even a topological manifold. Using the source's classification of this space as the octonionic isometry locus, we prove that a neighborhood of the identity is homeomorphic to a neighborhood of the origin in

$$\mathbb{R}^8 \times \{(A, B) \in \mathbb{R}^7 \times \mathbb{R}^7 : \text{rank}[A \ B] \leq 1\}.$$

The second factor is the rank-one determinantal cone. Its link is $(\mathbb{S}^6 \times \mathbb{S}^1)/\{\pm 1\}$, which has nonzero rational first homology. The resulting local homology is incompatible with a 16-manifold.

1 Source question and imported classification

A finite family $(x_1, \dots, x_k)$ in $\mathbb{O}^n$ is a weak associative orthonormal set when it is orthonormal and

$$\langle px_i, x_j \rangle - p\langle x_i, x_j \rangle = 0 \quad (p \in \mathbb{O})$$

for every i, j . The source defines

$$V_k^w(\mathbb{O}^n) = \{(x_1, \dots, x_k) : (x_1, \dots, x_k) \text{ is a weak associative orthonormal set}\}$$

and asks:

Is $V_k^w(\mathbb{O}^n)$ a manifold?

See Figure 1. Its modified projection question is answered inside the same paper, but this manifold question is left open.

We use one result from [1]. Regard a pair in $V_2^w(\mathbb{O}^2)$ as the two rows of a 2×2 octonionic matrix. Source Theorems 3.12 and 3.16 identify

$$V_2^w(\mathbb{O}^2) = \text{Iso}_{\mathbb{O}}(2) = \bigcup_{J \in \mathbb{S}^6} \mathbb{S}^7 U_{\mathbb{C}_J}(2), \quad \mathbb{C}_J = \mathbb{R} \oplus \mathbb{R}J.$$

Equivalently, every such matrix is pU for a unit octonion p and a unitary 2×2 matrix U over some complex slice $\mathbb{C}_J$; see Figure 2.

2 Proof intuition

Near the identity, the upper-left entry is nonzero. Dividing out its unit octonionic phase uniquely separates seven ordinary spherical directions. The remaining normalized matrix is complex-unitary in some slice $\mathbb{C}_J$ and has positive-real upper-left entry.

Such a unitary matrix has one real off-diagonal coordinate s , while its two imaginary parameters A and B must point in the same imaginary octonionic direction J . Thus the singular part of the local model is exactly the variety of collinear pairs in $\mathbb{R}^7$, or equivalently the 7×2 matrices of rank at most one. At the cone vertex, the direction J is undefined. The link remembers this collapse and has extra first homology, preventing local Euclidean structure.

3 The local normal form

Put

$$\mathcal{R} = \{(A, B) \in \mathbb{R}^7 \times \mathbb{R}^7 : A \wedge B = 0\} = \{(A, B) : \text{rank}[A \ B] \leq 1\}.$$

We identify $\mathbb{R}^7$ with the imaginary octonions.

Lemma 1 (Local determinantal chart). *Let I be the identity matrix in $V_2^w(\mathbb{O}^2)$. A neighborhood of I is homeomorphic, as a pointed space, to a neighborhood of 0 in $\mathbb{R}^8 \times \mathcal{R}$.*

Proof. Write a matrix near I as

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Its upper-left entry a is nonzero. Define

$$p = \frac{a}{|a|} \in \mathbb{S}^7, \quad U = \bar{p}, M,$$

where $\bar{p}$ multiplies every entry on the left. The proof of source Theorem 3.16 shows that $U \in U_{\mathbb{C}_J}(2)$ for some J , and its upper-left entry is $|a| > 0$. Conversely, pU is in the weak Stiefel space for every such pair. This normalization is unique, continuous, and has continuous inverse: the upper-left entry of pU is pr with $r > 0$, so it recovers p . Hence the germ splits as a neighborhood of 1 in $\mathbb{S}^7$ times the germ of normalized matrices U .

We now parametrize the normalized germ. A complex unitary matrix near I , with positive-real upper-left entry, has the unique form

$$U = \begin{pmatrix} r & s + B \\ -e^A(s - B) & e^A r \end{pmatrix}, \quad r = \sqrt{1 - s^2 - |B|^2}, \quad (1)$$

where $s \in \mathbb{R}$ is small and $A, B \in \mathbb{R}J$ for the same imaginary unit J . Indeed, the first row is (r, z) , with $z = s + B$, and every unit row orthogonal to it is $e^A(-\bar{z}, r)$, for a unique unit complex phase e^A near one.

The condition that A and B belong to one line $\mathbb{R}J$ is precisely $(A, B) \in \mathcal{R}$. Formula (1) is independent of a choice of J when one or both vectors vanish. Conversely, from U one recovers

$$s = \text{Re } U_{12}, \quad B = \text{Im } U_{12}, \quad A = \log(U_{22}/r),$$

using the unique logarithm near one. Thus the normalized germ is homeomorphic to the germ of $\mathbb{R} \times \mathcal{R}$. Finally, a neighborhood in $\mathbb{S}^7$ is $\mathbb{R}^7$, giving

$$(\text{Iso}_{\mathbb{O}}(2), I) \cong (\mathbb{R}^7 \times \mathbb{R} \times \mathcal{R}, 0).$$

□

4 The homology obstruction

Theorem 1. *The space $V_2^w(\mathbb{O}^2)$ is not a topological manifold. Consequently modified James Question (2) in [1] has a negative answer in general.*

Proof. The determinantal variety $\mathcal{R}$ is the cone on its unit link

$$L = \mathcal{R} \cap \mathbb{S}^{13} \cong (\mathbb{S}^6 \times \mathbb{S}^1) / \sim, \quad (u, v) \sim (-u, -v).$$

Indeed, every nonzero rank-one 7×2 matrix of Frobenius norm one has a factorization $uv^\top$ with $u \in \mathbb{S}^6$ and $v \in \mathbb{S}^1$, unique up to simultaneous sign.

The displayed quotient is a two-sheeted cover. With rational coefficients, transfer identifies its homology with the invariant part of $H_*(\mathbb{S}^6 \times \mathbb{S}^1; \mathbb{Q})$. The deck transformation is antipodal on both factors. On $H_1(\mathbb{S}^1; \mathbb{Q})$ it is rotation by π , hence has degree $+1$. Therefore

$$H_1(L; \mathbb{Q}) \cong \mathbb{Q}. \tag{2}$$

At a nonzero rank-one point, $\mathcal{R}$ is a smooth determinantal variety of dimension

$$1(7 + 2 - 1) = 8.$$

Lemma 1 therefore gives nearby smooth points of local dimension $8 + 8 = 16$. If a neighborhood of I were a manifold, its dimension would be 16.

For the pointed local model $X = \mathbb{R}^8 \times C(L)$, the local Kunneth formula and the cone formula give

$$H_k(X, X \setminus \{0\}; \mathbb{Q}) \cong \tilde{H}_{k-9}(L; \mathbb{Q}).$$

By (2), the local homology group in degree $k = 10$ is nonzero. A topological 16-manifold has local rational homology only in degree 16. Hence the identity is not locally Euclidean, proving the claim. $\square$

5 Verification, novelty, and scope

The exact checker verifies 21,000 rank-one minors and 60,000 rational complex-unitary and recovery identities, including exact recovery of the parameters in (1). These computations guard against coordinate and sign errors but do not replace the topological proof.

Four lightweight run indexes and bounded exact-title, weak-Stiefel, and octonionic-manifold searches found no duplicate or later resolution through 11 August 2026. The result answers the source’s general yes/no question negatively via $(k, n) = (2, 2)$. It does not classify which other pairs (k, n) may yield manifolds. Human review should focus on the uniqueness of the top-left normalization, the local parametrization (1), and the local-homology degree shift.

References

- [1] Q. Huo, G. Ren, and Z. Xu, *Octonionic isometric isomorphisms and partial isometry*, arXiv:2605.07226 (2026), Theorems 3.12 and 3.16 and Section 4.

octonionic Stiefel space:

- (1) When is the canonical projection

$$\pi : V_k(\mathbb{O}^n) \rightarrow V_l(\mathbb{O}^n) \quad (1 \leq l < k \leq n)$$

a fiber map?

- (2) Is $V_k(\mathbb{O}^n)$ a manifold?

However, in the sense of James given by (4.1), there may exist the space $V_3(\mathbb{O}^2)$ and $V_4(\mathbb{O}^2)$, examples can be found in [12, 22]. We think it is natural to define the **octonionic Stiefel spaces** as

$$V_k^w(\mathbb{O}^n) := \{(x_1, \dots, x_k) \mid x_i \in \mathbb{O}^n, (x_1, \dots, x_k) \text{ is a weak associative orthonormal set}\}.$$

Since the number of a weak associative orthonormal basis of $\mathbb{O}^n$ must be n , it follows that the space $V_k^w(\mathbb{O}^n)$ does not exist for $k > n$. The **James Questions** are canonically modified as

- (1) When is the canonical projection

$$\pi : V_k^w(\mathbb{O}^n) \rightarrow V_l^w(\mathbb{O}^n), (1 \leq l < k \leq n)$$

a fiber map?

- (2) Is $V_k^w(\mathbb{O}^n)$ a manifold?

Clearly, $V_1^w(\mathbb{O}^n)$ is the sphere $\mathbb{S}^{8n-1}$. However, $V_2^w(\mathbb{O}^n)$ is somewhat complicated. It turns out that the canonical projection

Figure 1: Source PDF page 23: the modified James questions, including whether $V_k^w(\mathbb{O}^n)$ is a manifold.

Theorem 3.16. $\text{Iso}_{\mathbb{O}}(2) \cong \bigcup_{J \in \mathbb{S}^6} \mathbb{S}^7 \times U_{\mathbb{C}_J}(2) / \sim$, where $\sim$ is an equivalence relation defined by

$$(p, \mathbf{T}) \sim (q, \mathbf{S}) \iff p\mathbf{T} = q\mathbf{S}.$$

Here $\mathbb{S}^6$ is the set of imaginary units of $\mathbb{O}$ and $U_{\mathbb{C}_J}(2) := U(\mathbb{C}_J^2)$ is the unitary group of $\mathbb{C}_J^2$.

Proof. By Theorem 3.12, we know that

$$\text{Iso}_{\mathbb{O}}(2) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : ((a, b), (c, d)) \text{ is a weak associative orthonormal basis of } \mathbb{O}^2 \right\}.$$

Suppose $a \in \mathbb{R}$. Since $A_p((a, b), (c, d)) = 0$ for all $p \in \mathbb{O}$, it follows from $a \in \mathbb{R}$ that

$$[p, c, d] = 0$$

for all $p \in \mathbb{O}$. This implies that there exists an imaginary unit J such that $c, d \in \mathbb{C}_J$. Using $(a, b) \perp (c, d)$ we obtain that

Figure 2: Source PDF page 22: the 2×2 octonionic isometry classification used in the local chart.